

Performance Testing

Date	30 June 2026
Team ID	xxxxxx
Project Name	Rising Waters Flood Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter, Locust, Postman
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	Flood Prediction API (/api/predict), Weather Data API (/api/weather), Alert Generation API (/api/alerts), User Authentication API (/api/auth/login)
Test Environment	AWS EC2 (t3.medium) - Staging Environment
Test Date	30 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal Load Test - Simulate typical user activity with mixed API calls (70% weather data, 20% predictions, 10% alerts)	100	300	All APIs respond within 2 seconds, error rate < 1%
2	Peak Load Test - Simulate heavy traffic during monsoon	500	300	Prediction API responds within 3 seconds, system remains stable

	season when users access predictions simultaneously			
3	Spike Test - Simulate sudden traffic spike (e.g., flash flood warning triggers mass access)	1000 (ramp-up in 30s)	180	System handles spike without crashing, auto-scaling triggers
4	Stress Test - Push system beyond breaking point to identify maximum capacity	2000	600	Identify breaking point and failure recovery time
5	Endurance Test - Sustained load for longer duration to detect memory leaks	200	3600	No memory leaks, stable performance over time
6	Database Load Test - Heavy read/write operations on TimescaleDB for weather data ingestion	1003	300	Database handles concurrent writes without performance degradation



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Well within acceptable limit
2	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	Peak response time acceptable
3	Throughput (Req/sec)	> 100 req/sec	185 req/sec	Pass	Handles high request volume efficiently
4	Error Rate	< 1%	0.2%	Pass	Minimal errors, mostly timeout-related
5	CPU Utilization	< 80%	65%	Pass	CPU usage stable under load
6	Memory Utilization	< 80%	58%	Pass	Memory usage stable, no leaks detected

Step 4: Observations & Analysis

S.No	Observations	Analysis
1	Excellent Performance	System performed well under normal and peak loads, exceeding target metrics across all scenarios
2	Scalability Confirmed	Auto-scaling triggered effectively at 500+ concurrent users, maintaining response times
3	ML Inference Speed	Model prediction API showed fast response times (avg 1.8s) even under high load
4	Alert Delivery	TimescaleDB handled high-volume time-series data with query times under 500ms
5	Real-time Updates	WebSocket connections maintained stable performance with minimal latency